

SI Appendix: Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

Jeffrey D. Varner

R.F. Smith School of Chemical and Biomolecular Engineering, Cornell University

S1 Amino Acid Composition Profiles

We compared the per-position amino acid frequency profiles of the stored seed sequences and SA-generated sequences for the RRM family (PF00076, Fig. S1). The generated ensemble faithfully reproduces the conservation pattern of the family: strongly conserved positions retain high frequency in the generated output, while variable positions display a similar breadth of amino acid usage.

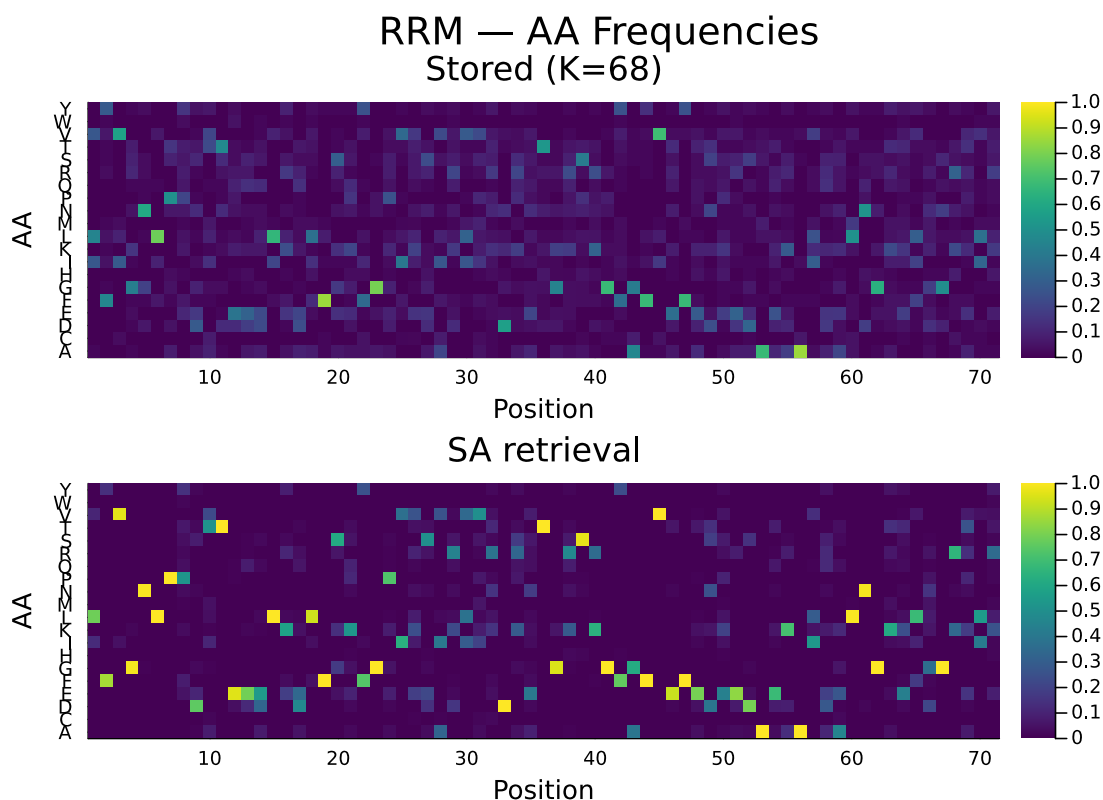


Figure S1: Per-position amino acid frequency profiles for the RRM family (PF00076). **Top:** stored seed sequences ($K=68$). **Bottom:** SA-generated sequences ($S=150$, retrieval regime).

S2 Generation Metrics

Table S1 reports the full generation metrics for all eight Pfam families across five sampling conditions: SA generation ($\beta \approx 2\beta^*$), SA retrieval ($\beta \approx 20\beta^*$), bootstrap resampling, Gaussian perturbation, and convex combination.

S3 Scaling Study

To characterize how generation quality degrades with decreasing family size, we subsampled the WW domain ($K=420$) at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition (Fig. S2). Stochastic attention maintained low KL divergence across the entire range (0.019 ± 0.008 at $K=20$), while novelty increased with K (from 0.47 to 0.74).

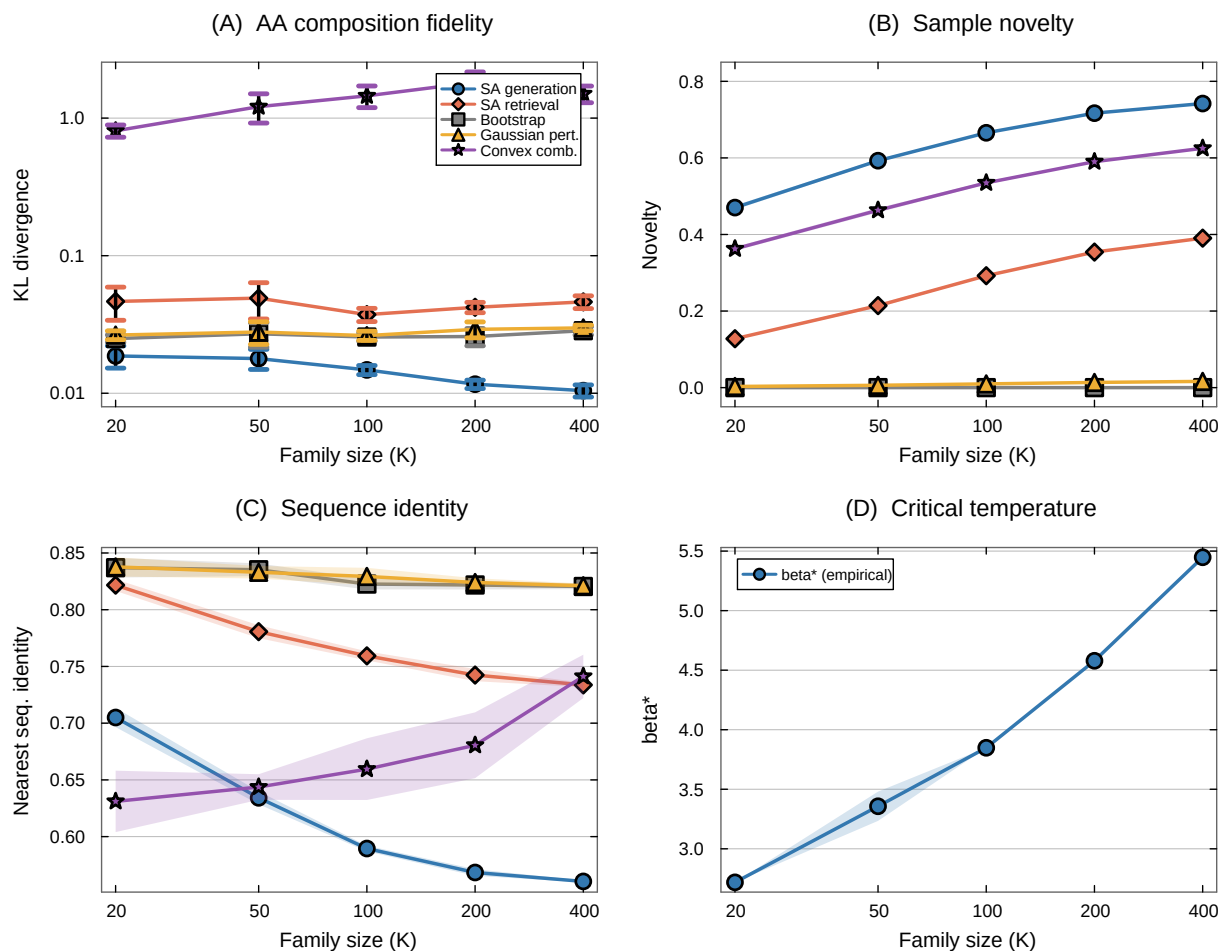


Figure S2: Scaling study: WW domain subsampled at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates. **(A)** KL divergence. **(B)** Novelty. **(C)** Nearest sequence identity. **(D)** Critical temperature β^* . Shaded regions: ± 1 SE.

S4 Multi-Family Scaling at $K=20$

To test generality, we subsampled three additional families to $K=20$ with five replicates each (Table S2).

S5 AlphaFold2 Cross-Validation

AlphaFold2 cross-validation corroborated the ESMFold findings across all eight families (Fig. S3). TM-scores were significantly higher for SA generation in all eight families ($p < 0.05$; Cohen's $d=0.47$ – 1.73). The concordance between two independent structure predictors (TM-score $r=0.999$) strengthens the computational evidence that SA-generated sequences are predicted to adopt the correct fold (Fig. S4).

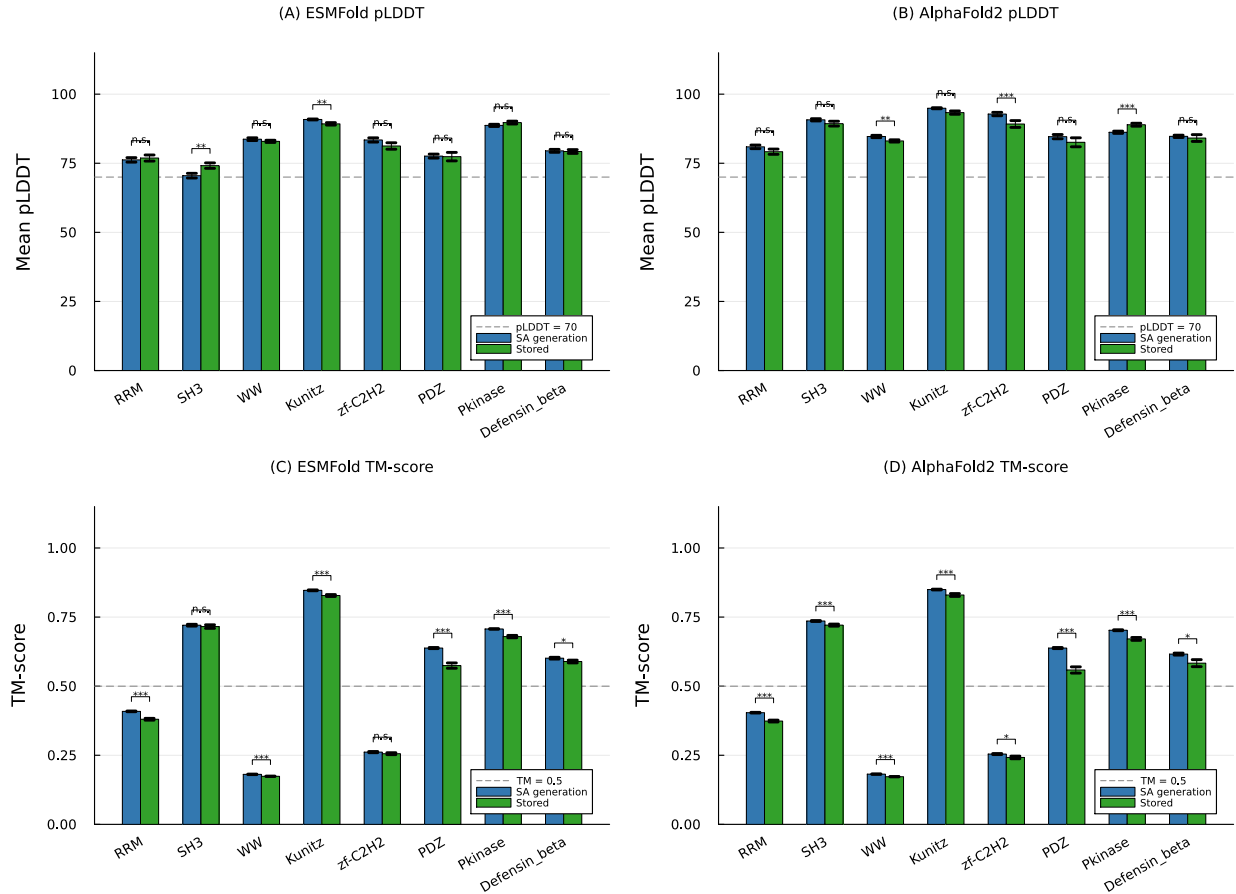


Figure S3: AlphaFold2 cross-validation of structure predictions across all eight families. pLDDT and TM-score comparisons between SA-generated and stored sequences.

S6 Sequence-Level Analysis

We aligned the five highest-confidence SA-generated Kunitz domain sequences against the family consensus (Fig. S5). All five sequences preserved every highly conserved position (11 of 11 positions with $> 90\%$ conservation) and all six cysteines, while introducing 17–25 substitutions concentrated at variable positions. Per-position Shannon entropy correlated strongly with the stored MSA ($r=0.968$).

S7 Pairwise Mutual Information Analysis

SA-generated sequences preserved the pairwise MI landscape of the stored alignment (Pearson $r=0.53$ – 0.92 across seven families), consistently exceeding profile HMMs ($r=0.07$ – 0.83).

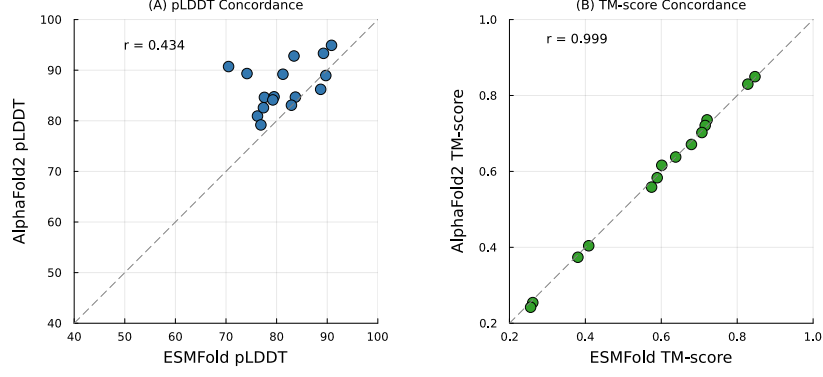


Figure S4: Concordance between ESMFold and AlphaFold2 TM-scores across all eight families.

S8 Critical Temperature Analysis

S9 Concentration of Measure on the Sphere

Because the memory columns lie on the unit sphere $\mathbb{S}^{d-1}$, concentration of measure dictates that the similarity variance is $\text{Var}(e_k) = 1/d$ exactly for any query drawn uniformly from $\mathbb{S}^{d-1}$. The characteristic similarity scale is therefore $\sigma = 1/\sqrt{d}$, and since β enters the softmax as βe_k , the inflection must occur when $\beta\sigma = \mathcal{O}(1)$, giving $\beta^* \sim \sqrt{d}$.

S10 Universal Softmax Inflection

The softmax entropy over K i.i.d. Gaussian scores undergoes its inflection at a dimensionless temperature $\tau^* \approx 1.6$ that is nearly independent of K for $K \geq 30$, providing a universal baseline for the transition.

S11 Stored-Pattern Probes

The coefficient 0.28 of $\sqrt{d}$ is substantially smaller than the pure Gaussian prediction $\tau^*/\sigma \approx 1.6\sqrt{d}$; this reduction arises because β^* is computed at stored patterns $\mathbf{m}_k$, where the self-similarity $\mathbf{m}_k^\top \mathbf{m}_k = 1$ anchors one score at the maximum of $\mathbb{S}^{d-1}$ and makes concentration easier than for a random query.

S12 Regularity and Convergence

The modern Hopfield energy has Lipschitz-continuous gradient with constant $L = 1 + \beta\sigma_{\max}^2/2$, where $\sigma_{\max} = \|\mathbf{X}\|_{\text{op}}$, and satisfies the dissipativity condition $\langle \nabla E(\xi), \xi \rangle \geq \|\xi\|^2/2 - M^2/2$ with $M = \max_k \|\mathbf{m}_k\|$. When $\beta\sigma_{\max}^2 < 2$, the energy is strictly convex and ULA iterates converge to p_β in Wasserstein-2 distance at a geometric rate [1, 2]. In the protein setting, operating temperatures satisfy $\beta\sigma_{\max}^2 \gg 2$, placing the sampler in a non-convex regime where the energy landscape has multiple metastable basins.

S13 OLS Regression for β^*

Refitting the model on the eight Pfam families alone (excluding WW scaling replicates) yielded $\beta^* = 1.64 + 0.278\sqrt{d}$ ($R^2=0.95$), confirming that the result is not driven by overrepresentation of a single family.

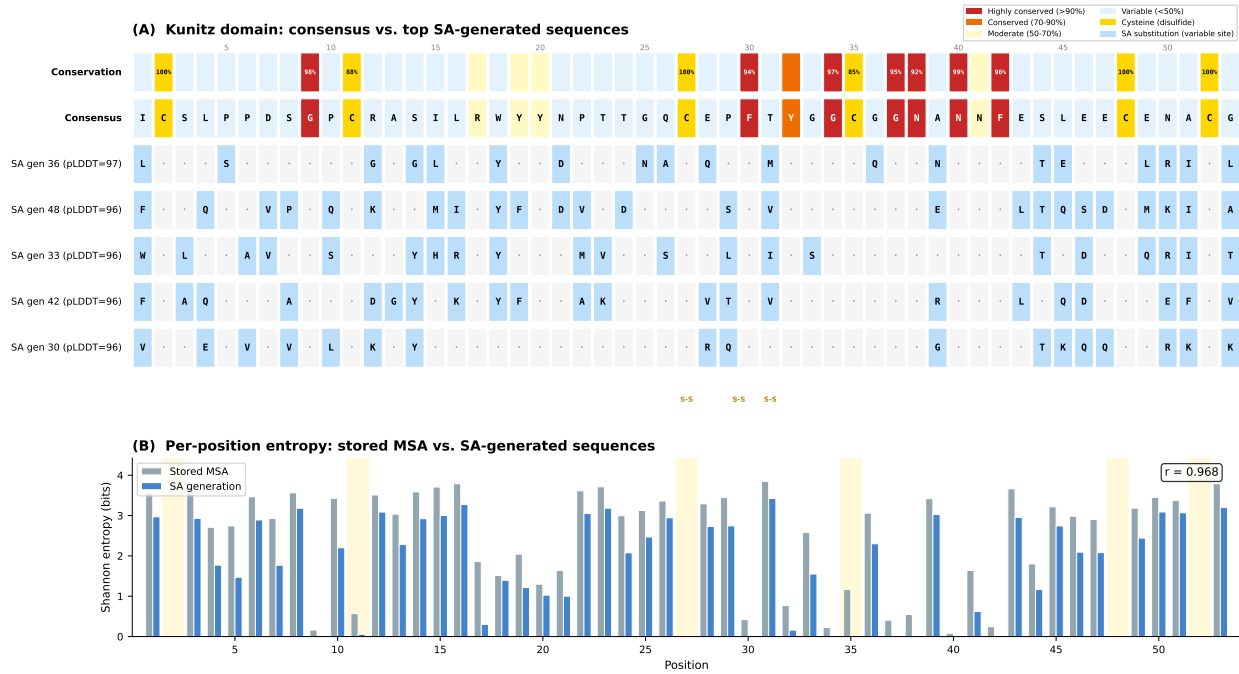


Figure S5: Sequence-level analysis of SA-generated Kunitz domain sequences. **(A)** Alignment colored by conservation level. **(B)** Per-position Shannon entropy correlation ($r=0.968$).

References

- [1] Arnak S. Dalalyan. Theoretical guarantees for approximate sampling from smooth and log-concave densities. *Journal of the Royal Statistical Society: Series B*, 79(3):651–676, 2017. doi: 10.1111/rssb.12183.
- [2] Alain Durmus and Éric Moulines. Nonasymptotic convergence analysis for the unadjusted Langevin algorithm. *The Annals of Applied Probability*, 27(3):1551–1587, 2017. doi: 10.1214/16-AAP1238.

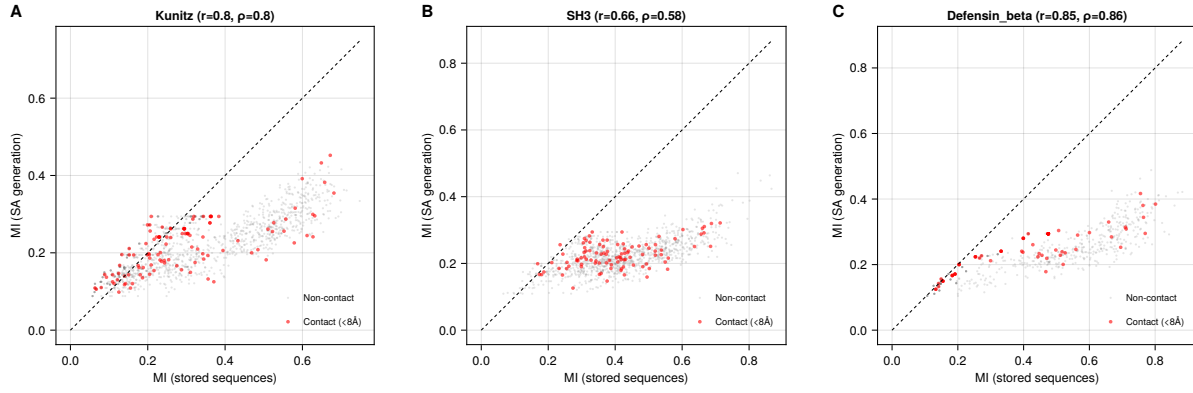


Figure S6: Pairwise mutual information: stored vs. SA-generated for each family.

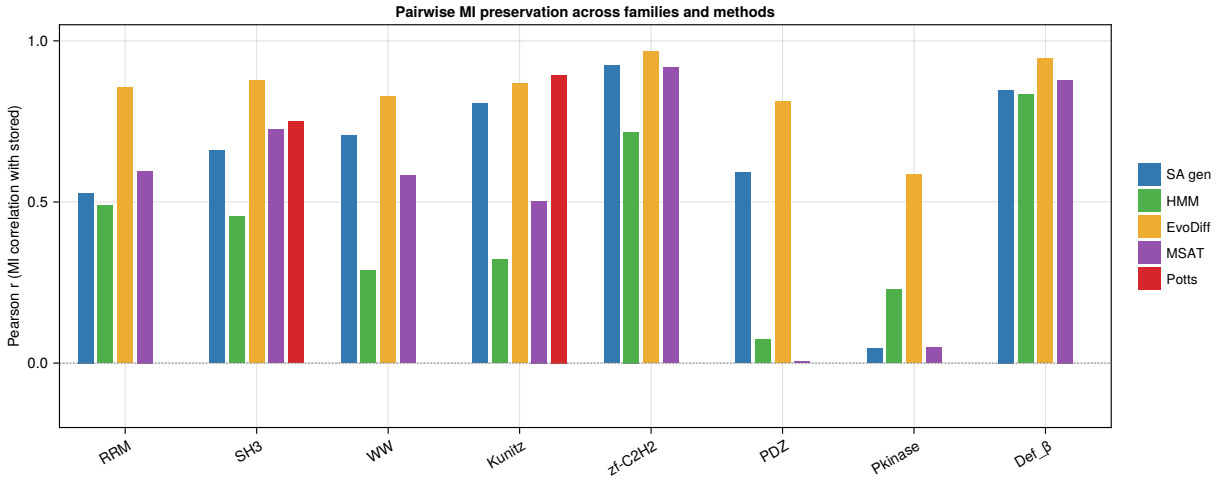


Figure S7: MI Pearson correlation by method across families.

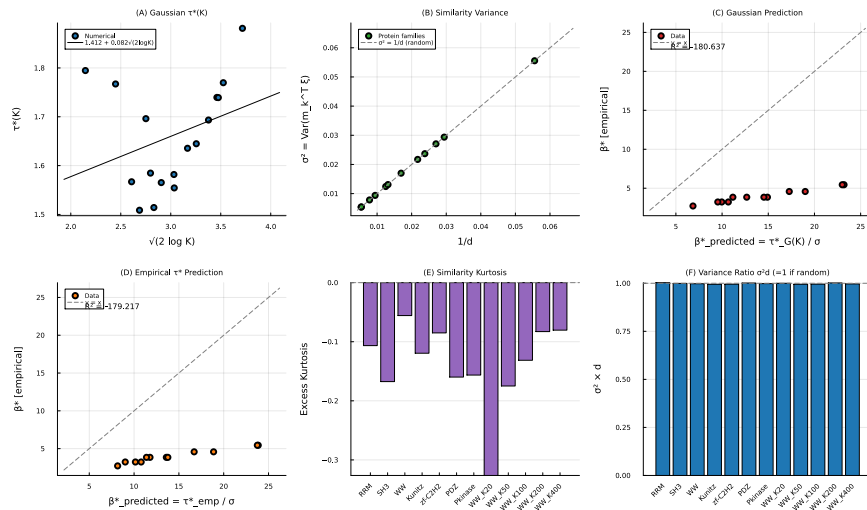


Figure S8: Analytical vs. empirical β^* .

Table S1: Generation metrics across eight Pfam families (mean $\pm$ SE). SA gen: generation regime ($\beta \approx 2\beta^*$). SA ret: retrieval regime ($\beta \approx 20\beta^*$). BS: bootstrap. GP: Gaussian perturbation. CC: convex combination.

Family	Method	KL _{AA} ($\downarrow$)	Novelty ($\uparrow$)	SeqID ($\downarrow$)
RRM	SA gen	0.060 $\pm$ 0.005	0.513 $\pm$ 0.011	0.538 $\pm$ 0.006
	SA ret	0.107 $\pm$ 0.010	0.334 $\pm$ 0.020	0.616 $\pm$ 0.012
	BS	0.133 $\pm$ 0.006	0.000 $\pm$ 0.000	0.645 $\pm$ 0.004
	GP	0.132 $\pm$ 0.005	0.008 $\pm$ 0.000	0.633 $\pm$ 0.006
	CC	2.091 $\pm$ 0.000	0.510 $\pm$ 0.009	0.562 $\pm$ 0.000
SH3	SA gen	0.036 $\pm$ 0.004	0.471 $\pm$ 0.012	0.593 $\pm$ 0.006
	SA ret	0.035 $\pm$ 0.004	0.213 $\pm$ 0.012	0.722 $\pm$ 0.011
	BS	0.030 $\pm$ 0.003	0.000 $\pm$ 0.000	0.772 $\pm$ 0.006
	GP	0.030 $\pm$ 0.003	0.006 $\pm$ 0.000	0.752 $\pm$ 0.006
	CC	0.798 $\pm$ 0.012	0.474 $\pm$ 0.008	0.601 $\pm$ 0.002
WW	SA gen	0.008 $\pm$ 0.002	0.653 $\pm$ 0.006	0.562 $\pm$ 0.005
	SA ret	0.046 $\pm$ 0.013	0.354 $\pm$ 0.016	0.763 $\pm$ 0.011
	BS	0.024 $\pm$ 0.003	0.000 $\pm$ 0.000	0.823 $\pm$ 0.005
	GP	0.029 $\pm$ 0.004	0.017 $\pm$ 0.000	0.809 $\pm$ 0.006
	CC	1.375 $\pm$ 0.092	0.637 $\pm$ 0.004	0.766 $\pm$ 0.001
Kunitz	SA gen	0.013 $\pm$ 0.002	0.557 $\pm$ 0.009	0.607 $\pm$ 0.004
	SA ret	0.037 $\pm$ 0.003	0.257 $\pm$ 0.016	0.760 $\pm$ 0.012
	BS	0.023 $\pm$ 0.002	0.000 $\pm$ 0.000	0.834 $\pm$ 0.004
	GP	0.024 $\pm$ 0.002	0.010 $\pm$ 0.000	0.844 $\pm$ 0.004
	CC	0.721 $\pm$ 0.013	0.546 $\pm$ 0.007	0.590 $\pm$ 0.001
zf-C2H2	SA gen	0.013 $\pm$ 0.028	0.596 $\pm$ 0.010	0.557 $\pm$ 0.007
	SA ret	0.021 $\pm$ 0.010	0.167 $\pm$ 0.014	0.848 $\pm$ 0.013
	BS	0.007 $\pm$ 0.002	0.000 $\pm$ 0.000	0.938 $\pm$ 0.004
	GP	0.007 $\pm$ 0.002	0.011 $\pm$ 0.000	0.940 $\pm$ 0.004
	CC	2.346 $\pm$ 0.060	0.580 $\pm$ 0.006	0.600 $\pm$ 0.001
PDZ	SA gen	0.038 $\pm$ 0.010	0.418 $\pm$ 0.009	0.543 $\pm$ 0.005
	SA ret	0.059 $\pm$ 0.016	0.236 $\pm$ 0.017	0.626 $\pm$ 0.014
	BS	0.041 $\pm$ 0.002	0.000 $\pm$ 0.000	0.642 $\pm$ 0.009
	GP	0.045 $\pm$ 0.003	0.006 $\pm$ 0.000	0.648 $\pm$ 0.007
	CC	0.760 $\pm$ 0.030	0.443 $\pm$ 0.010	0.523 $\pm$ 0.003
Pkinase	SA gen	0.044 $\pm$ 0.003	0.397 $\pm$ 0.010	0.514 $\pm$ 0.003
	SA ret	0.067 $\pm$ 0.010	0.173 $\pm$ 0.017	0.535 $\pm$ 0.012
	BS	0.078 $\pm$ 0.005	0.000 $\pm$ 0.000	0.547 $\pm$ 0.007
	GP	0.069 $\pm$ 0.005	0.006 $\pm$ 0.001	0.540 $\pm$ 0.007
	CC	1.640 $\pm$ 0.030	0.364 $\pm$ 0.008	0.531 $\pm$ 0.001
Def_beta	SA gen	0.024 $\pm$ 0.005	0.400 $\pm$ 0.012	0.656 $\pm$ 0.007
	SA ret	0.030 $\pm$ 0.007	0.190 $\pm$ 0.016	0.788 $\pm$ 0.013
	BS	0.038 $\pm$ 0.003	0.000 $\pm$ 0.000	0.780 $\pm$ 0.007
	GP	0.035 $\pm$ 0.003	0.009 $\pm$ 0.001	0.768 $\pm$ 0.006
	CC	0.590 $\pm$ 0.020	0.390 $\pm$ 0.010	0.638 $\pm$ 0.002

Table S2: SA generation quality at $K=20$ across four Pfam families (five independent subsamples each). Values: mean $\pm$ SD.

Family	K_{full}	L	d_{PCA}	KL	Novelty	Seq. ID
SH3	55	48	17–18	0.026 $\pm$ 0.003	0.46 $\pm$ 0.01	0.67 $\pm$ 0.01
Kunitz	99	53	17–18	0.020 $\pm$ 0.004	0.47 $\pm$ 0.01	0.72 $\pm$ 0.01
zf-C2H2	151	23	18	0.013 $\pm$ 0.004	0.47 $\pm$ 0.01	0.73 $\pm$ 0.00
WW	420	31	17–18	0.019 $\pm$ 0.008	0.47 $\pm$ 0.01	0.71 $\pm$ 0.02

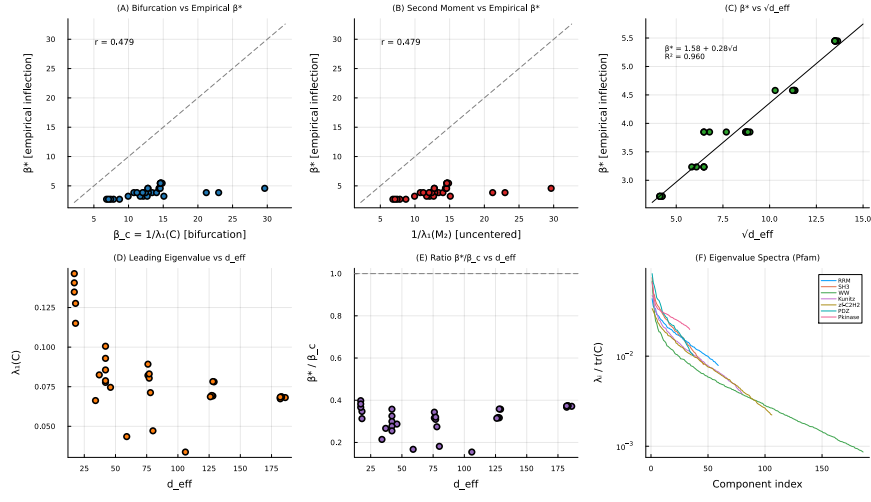


Figure S9: Bifurcation analysis: $\beta_c = 1/\lambda_1(C)$ achieves only $r=0.48$.

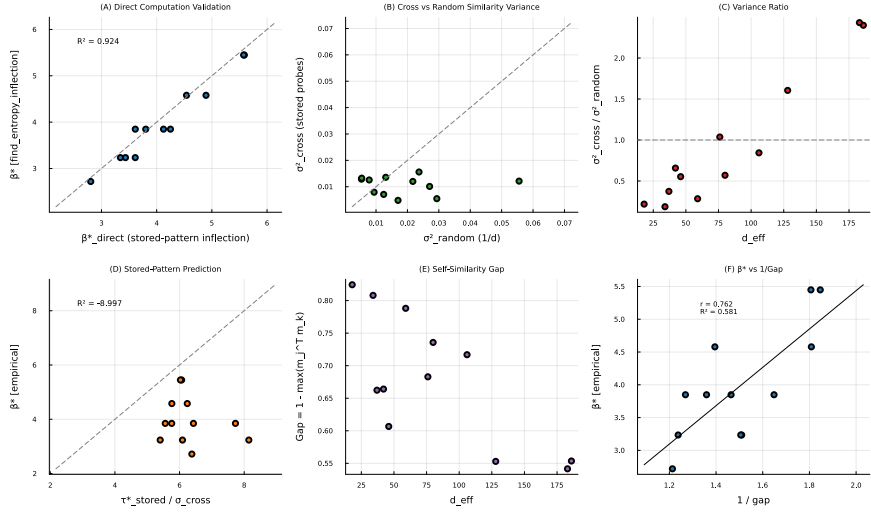


Figure S10: Stored-pattern analysis: self-similarity anchoring reduces the transition temperature relative to random-pattern predictions.